

AC 209b Milestone 2

Intro and Data Demo

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Project Motivation

- **Core question:** can emotion-related language help explain self-reported MBTI patterns in Reddit writing?
- **Why this project is interesting:**
 - emotion is more interpretable than a black-box personality classifier
 - personality labels are noisy, so we need to verify what the data can realistically support
 - this milestone is not just about prediction, but about using EDA to refine the research question
- **Our setup:** use one cleaner labeled dataset for emotion supervision and one large Reddit dataset with self-reported MBTI labels
- **Key assumption:** we are **not** assuming that one emotion directly maps to one personality type; instead, we ask whether **emotion-related language patterns** can help explain MBTI dimensions

Data Acquisition and Preparation

- **Emotion dataset:** 20,000 short texts, 6 balanced labels, 2.35 MB
- **Reddit MBTI dataset:** 13,028,455 posts, 11,773 authors, 16 self-reported types, 2.70 GB
- **Wrangling:**
 - standardized labels
 - removed 180 blank posts
 - derived 4 binary dimensions
 - used a 250,000-post sample for text-length EDA
- **Why this matters:**
 - consistent schema
 - binary dimensions are more realistic
 - cleaner baseline pipeline

EDA: Clean Source, Difficult Target

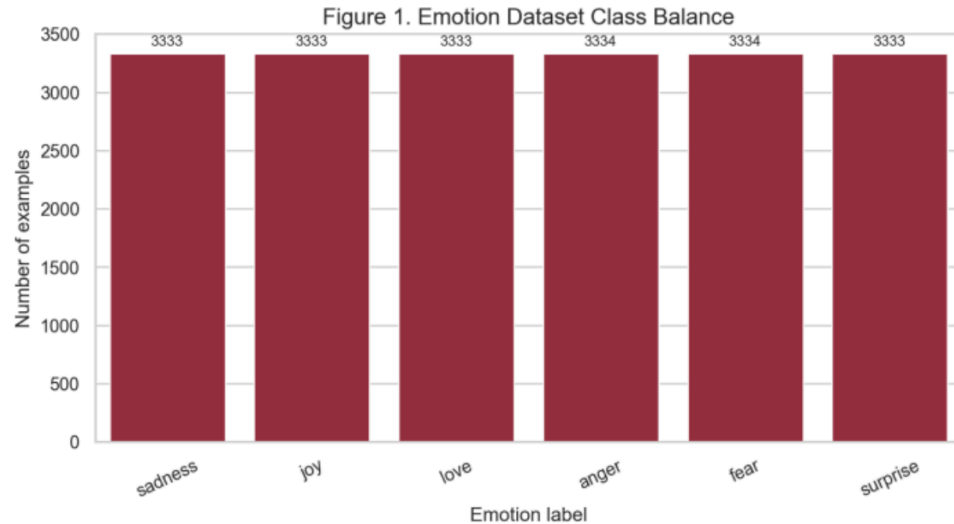


Figure 1: Emotion dataset class balance

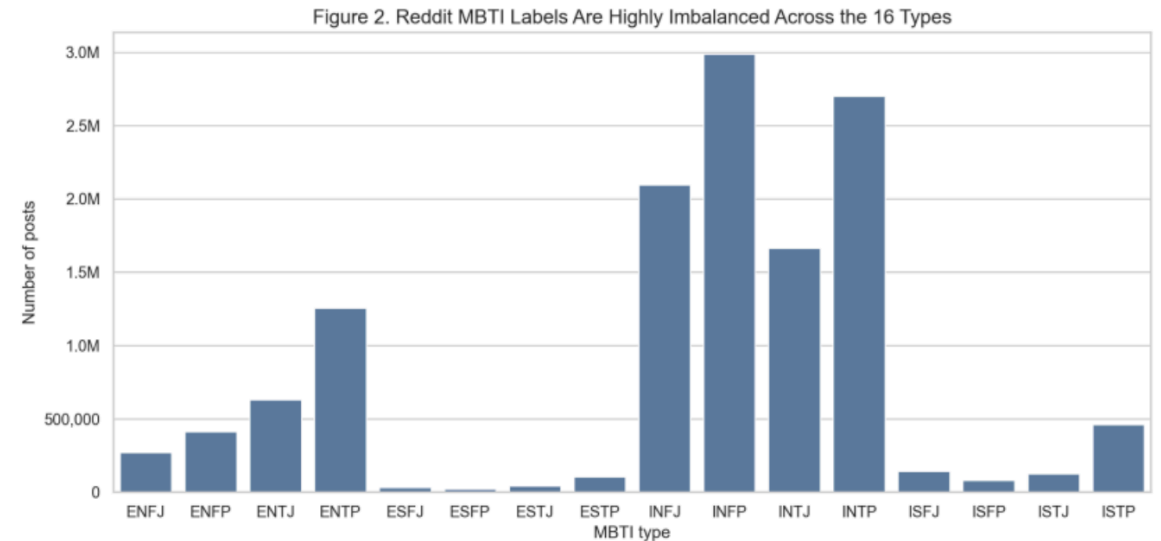


Figure 2: MBTI type imbalance

- The emotion source dataset is almost perfectly balanced, with about 3,333 examples per class.
- The MBTI target dataset is heavily skewed: INFP is **22.94%** of all posts, while ESFP is only **0.18%**.
- **Insight:** the source task is clean, but the target task is much harder.

EDA: Author Concentration Changes the Modeling Plan

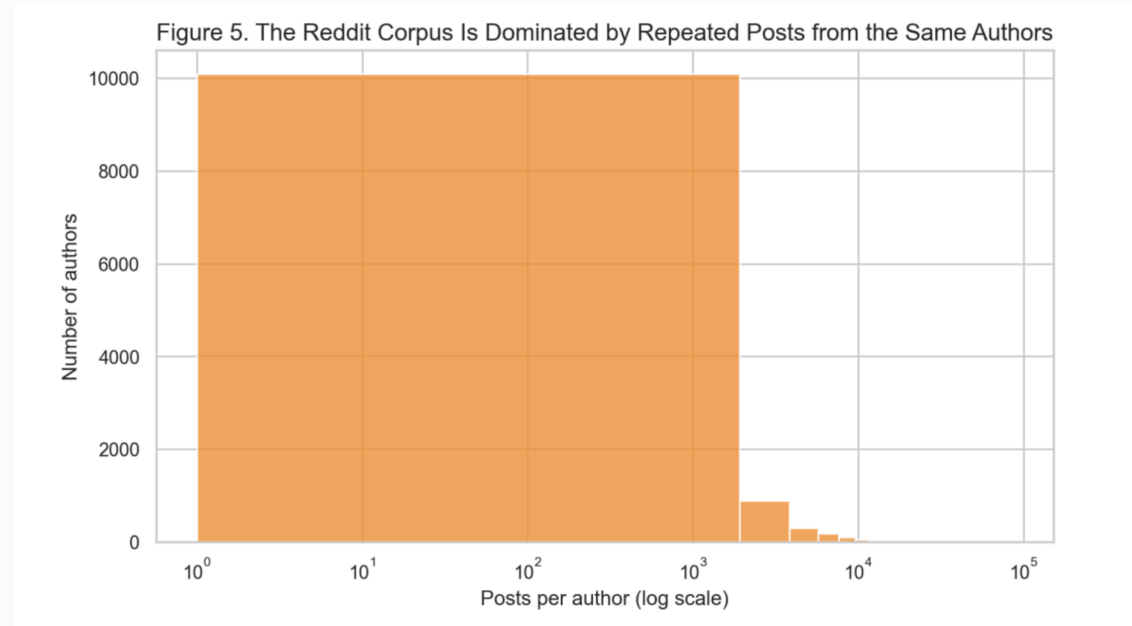


Figure 3: Posts per author in the Reddit MBTI dataset

- The median author contributes **272** posts; the 99th percentile contributes **12,753**.
- The top 1% of authors generate **18.98%** of all posts.
- **Main implication:** train/test splits must be done at the **author level**, not the post level.
- **Further implication:** post-level predictions alone may reflect user activity patterns rather than stable personality signals.

EDA: Why We Need to Rescope the Question

- **What the EDA suggests:**
 - 16-way MBTI prediction is difficult because the labels are highly imbalanced
 - MBTI labels are self-reported, so they are likely noisy
 - users contribute very different numbers of posts, so naive post-level modeling would be misleading
- **What this means for the project:**
 - the data is better suited for **dimension-level analysis** than for full-type classification
 - the more realistic unit of analysis is the **user**, not the single post
- **Resulting direction:** use EDA to narrow the scope before building more complex models

Redefinition and Rescoping

- **Original plan:** 16-type MBTI classification from Reddit text
- **EDA findings:**
 - extreme class imbalance
 - noisy self-reported labels
 - strong author-level clustering
- **Rescope:**
 - move to **4 binary dimensions** (E/I , N/S , F/T , J/P)
 - shift from post-level to **user-level analysis**
- **Revised question:**
 - can emotion-related language patterns help explain MBTI dimensions at the user level?
- **Reason for this redefinition:**
 - it is more aligned with what the data can support

Assumptions and Limitations

- We do **not** assume that a specific emotion directly maps to a specific MBTI type. Instead, we ask whether people with different personality dimensions may express emotion differently in text.
- Our downstream MBTI analysis depends on the quality of the first-stage emotion classifier, so the alignment between source emotion labels and text matters.
- Emotion cannot be fully captured by text alone, since tone, facial expression, and other nonverbal cues are missing.
- Because both datasets come from online text platforms, our findings are more relevant to **social media language** than to personality expression in general.

Baseline Models and Pipeline

- **Step 1: emotion classification baseline**
 - train a baseline text classifier on the emotion dataset
 - this gives us predicted emotion-related signals for Reddit posts
- **Step 2: MBTI dimension baseline**
 - build baseline models for the four MBTI binary dimensions
 - compare models that use basic text features with models that incorporate emotion-related features
- **Why this pipeline matters:**
 - the second step depends on the quality of the first step
 - if emotion predictions are too noisy, they may weaken downstream MBTI analysis
- **Main takeaway:**
 - this is a two-step pipeline, so evaluating the first stage carefully is essential

Next Steps

- **Emotion model**
 - build a baseline classifier on the emotion dataset
 - explore stronger encoder-based models if needed
- **Feature construction**
 - generate emotion scores for Reddit posts
 - aggregate them into **user-level profiles**
- **MBTI analysis**
 - predict the 4 binary dimensions
 - compare baseline and emotion-informed features
- **Evaluation**
 - use **author-level splits**

Future Considerations

- **Models under consideration**

- baseline text classifier for emotion
- logistic regression and random forest for downstream MBTI dimension prediction
- stronger encoder-based models as future extensions

- **Challenges**

- noisy self-reported MBTI labels
- possible mismatch between predicted emotion and true underlying emotion
- class imbalance and unequal user activity

- **Open questions**

- is the binary rescope sufficient for a meaningful result?
- how much does the first-stage emotion model help the second-stage MBTI task?
- how should we balance interpretability and predictive performance?